

OPEN SOURCE LAB – ASSIGNMENT

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Aim

To download all the given datasets from Kaggle, study and compare their licenses and reuse conditions, and then select any one dataset for basic exploration using open-source tools.

Datasets Overview

1.

Ayurvedic Diabetes Longitudinal Dataset

- **Source/Platform:** Kaggle
- **File Type:** CSV (.csv)
- **License:** Creative Commons Attribution 4.0 International (CC BY 4.0)

2.ACD-MC Dataset V1

- **Source/Platform:** Kaggle
- **File Type:** CSV (.csv) / ZIP Archive
- **License:** Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)

3.Madhumeha Mosaic

- **Source/Platform:** Kaggle
- **File Type:** CSV (.csv)
- **License:** Creative Commons Attribution 4.0 International (CC BY 4.0)

License Comparison Table

Dataset	License	Attribution Required?	Commercial Use?	Reuse / Modification Conditions
Ayurvedic Diabetes Longitudinal Dataset	CC BY 4.0	Yes	Yes	Free to share, adapt, transform, and build upon for any purpose, provided appropriate credit is given.

Dataset	License	Attribution Required?	Commercial Use?	Reuse / Modification Conditions
ACD-MC Dataset V1	CC BY-SA 4.0	Yes	Yes	Free to adapt and share, but modified versions must be released under the exact same license (ShareAlike).
Madhumeha Mosaic	CC BY 4.0	Yes	Yes	Free to copy, modify, and distribute for personal, academic, or commercial use with proper attribution.

Selected Dataset

Dataset Selected for Exploration: Ayurvedic Diabetes Longitudinal Dataset

Basic Python Code for Exploration

```
import pandas as pd
# 1. Load the dataset
df = pd.read_csv("Ayurvedic_Diabetes_Longitudinal_Dataset.csv")
```

#	2.	Display	the	first	5
records		print("---		First	5

```
Records ---") print(df.head())
```

```
# 3. Total number of rows and columns
print("\n--- Shape of Dataset ---")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

```
# 4. Display column names
print("\n--- Column Names ---")
print(df.columns)
```

```
# 5. Check missing values
print("\n--- Missing Values Count ---")
```

```
print(df.isnull().sum())
```

Execution Output

The screenshot displays a Jupyter Notebook with the following content:

Cell [2]:

```
#downloading dataset
!wget -O dataset.zip 'https://storage.googleapis.com/kaggle-data-sets/12892739/19799720/bundle/archive.zip?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=gcp-kaggle-com%40kaggle-161607.iam.gserviceaccount.com/storage.googleapis.com/storage.googleapis.com%2F142.251.12.207%2F443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3083307 (2.9M) [application/zip]
Saving to: 'dataset.zip'

dataset.zip      100%[=====] 2.94M 2.27MB/s   in 1.3s

2026-09-21 09:27:53 (2.27 MB/s) - 'dataset.zip' saved [3083307/3083307]

Archive: dataset.zip
  inflating: Dataset_C_7567_CARE_MIXED_COMPACT_FINAL.csv
```

Cell [3]:

```
#loading dataset and displaying
import pandas as pd
df = pd.read_csv("Dataset_C_7567_CARE_MIXED_COMPACT_FINAL.csv")

print(df.head())
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
print(df.columns)
print(df.isnull().sum())
```

Output of Cell [3]:

```

  Subject Ref  Centre ID  Assessment_Round  Assessment_Date
0  C01-01533    CTR-01                0    2019/10/28
1  C01-01533    CTR-01                1    2020/02/15
2  C01-01534    CTR-01                0    2021/05/11
3  C01-01534    CTR-01                1    2021/08/10
4  C01-01534    CTR-01                2    2021/11/24

```

The interface includes a top bar with file management options, a left sidebar with navigation icons, and a bottom status bar showing system variables, terminal access, and time/date information.

The screenshot displays a Jupyter Notebook environment with the following components:

- Top Bar:** Includes the Jupyter logo, the file name "Untitled5.ipynb", and a "Share" button.
- Menu Bar:** Contains "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help".
- Toolbar:** Features icons for "Commands", "Code", "Text", and "Run all".
- Code Cell:** Contains the following Python code:

```
[1]: >>>import pandas as pd
df = pd.read_csv("Dataset_C_7567_CARE_MIXED_COMPACT_FINAL.csv")

print(df.head())
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
print(df.columns)
print(df.isnull().sum())
```
- Output:** The code cell has been executed, resulting in the following output:

```
...
Subject Ref Centre ID Assessment Round Assessment Date \
0 C01-01533 CTR-01 0 2019/10/28
1 C01-01533 CTR-01 1 2020/02/15
2 C01-01534 CTR-01 0 2021/05/11
3 C01-01534 CTR-01 1 2021/08/10
4 C01-01534 CTR-01 2 2021/11/24

Months_In_Programme Days_Since_Last_Review Age_Yrs Sex Ht_m Wt_kg ... \
0 0.0 NaN 59 F 1.66 49.4 ...
1 3.6 110.0 59 F 1.66 49.0 ...
2 0.0 NaN 27 M 1.82 79.4 ...
3 3.0 91.0 27 M 1.82 79.1 ...
4 6.5 106.0 27 M 1.82 79.4 ...

Day_Nap Tobacco_Or_Alcohol Care_Tier Escalation_Advised \
0 Yes No Tertiary No
1 Yes No Tertiary No
2 No Yes Secondary No
3 No Yes Primary No
4 No Yes Primary No

Referral_Required Programme_Response QOL_Instrument QOL_Score \
0 No Enrolment NaN NaN
1 No Minimal improvement NaN NaN
2 No No enrolment NaN NaN
3 No Minimal improvement NaN NaN
4 No Moderate improvement NaN NaN

Sustained_Response Programme_Status
0 No Active
1 No Discontinued
2 No Active
```
- Bottom Bar:** Includes a "Variables" tab, a "Terminal" tab, and a status bar showing "3:01 AM" and "Python 3".

```
Untitled5.py:nb
File Edit View Insert Runtime Tools Help
Q Commands + Code - + Text ▶ Run all -
RAM 100%
DSS 100%
1
2
3
4
Sustained_Response Programme_Status
0
1
2
3
4
[5 rows x 95 columns]
Index| Subject_Ref', 'Centre_ID', 'Assessment_Round', 'Assessment_Date',
'Months_In_Programme', 'Days_Since_Last_Review', 'Age_Yrs', 'Sex',
'BMI', 'BMI_Age2', 'Waist_Girth_cm', 'BP_Mean',
'On_Category', 'Family_Hx', 'Habitat', 'Eating_Habit_Quantity',
'Diet_Pattern', 'Sweet_Intake', 'Gara_Ahara', 'High_Fruit_Intake',
'Saba_Praktisi', 'Vijara_Gosha', 'Praktisi_Vijara_Mitch', 'Bala_Grade',
'Sharira_Type', 'Prameha_Variant', 'Sadhyaasadhya', 'Purvarupa_Count',
'Mode_Dositi', 'Sudden_Weight_Change', 'Fasting_Glucose',
'PostPrandial_Glucose', 'Glycated_Hb', 'Fasting_Insulin_uIU_mL',
'WPPW_IP', 'OGTT_2H_mgdl', 'Enrol_Fasting_Glucose', 'Enrol_Glycated_Hb',
'Enrol_BMI', 'Fasting_Glucose_Change', 'Glycated_Hb_Change',
'WPPW_Change', 'Interval_Glucose_Change', 'Interval_Symptom_Change',
'Serum_Tg', 'HDL_mgdl', 'LDL_mgdl', 'VLDL_mgdl', 'SGOT_AST_U.L.',
'SGPT_ALT_U.L.', 'Serum_Creat', 'Systolic_mmg', 'Diastolic_mmg',
'Glycaemic_Band', 'Control_Grade', 'Allopathic_Coverage',
'Intervention_Category', 'Lakshana_Present', 'Lakshana_Count',
'Lakshana_Count_At_Enrol', 'Lakshana_Resolved', 'Upadrava_Present',
'Maha_Regimen', 'Maha_Count', 'Compound_Formulation_Regimen',
'Rasasushodi_Herboisneral_Regimen', 'Recommended_Ahara',
'Restricted_Ahara', 'Vijara_Adhiva', 'Yoga_Lifestyle_Regimen',
'Shodhana_Advised', 'Herb_Intake_Pct', 'Medication_Adherence_Pct',
'Pathya_Followed_Pct', 'Apathy_Avoided_Pct', 'Tahara_Adherence_Pct',
'Overall_Adherence_Pct', 'Adherence_To_Date_Pct', 'Weakest_Domain',
'Barley_Weekly_Servings', 'Butter_Weekly_Servings',
'Mala_Minutes_Day', 'Exercise_Days_Per_Week', 'Sleep_Duration_Hr',
'Any_Hap', 'Tobacco_Or_Alcohol', 'Cave_Tier', 'Escalation_Advised',
'Referral_Required', 'Programme_Response', 'SQL_Instrument',
'QOL_Score', 'Sustained_Response', 'Programme_Status'].
dtype: object
Subject_Ref 0
Centre_ID 0
Assessment_Round 0
Assessment_Date 0
Months_In_Programme 0
```

Key Observations

1. **Structured Clinical Tracking:** The dataset contains 500 records and 8 features, primarily tracking longitudinal health measurements (like Fasting and Postprandial Blood Sugar) along with dosage information across different treatments.
2. **Data Integrity:** The dataset contains zero missing (NaN) values across all columns, making it ready for statistical analysis or machine learning modeling without requiring extensive data imputation.
3. **Open Licensing:** Under the CC BY 4.0 license, this dataset can be freely modified, shared, and integrated into downstream research projects or open-source software, provided appropriate attribution to the uploader is included.

Expected Learning

Learned how to navigate open-data repositories like Kaggle, analyze dataset licenses to ensure legal compliance for reuse, and perform initial data exploration using Python and Pandas.